

COLUMBUS, NE, USA

WMO: 725565

Lat: 41.450N

Lon: 97.345W

Elev: 441

StdP: 96.14

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94951

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.7	-17.7	-24.9	0.4	-19.9	-22.4	0.5	-17.2	15.3	-5.4	13.4	-5.7	3.4	330	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	34.0	23.8	32.6	23.3	31.3	22.8	25.7	31.7	24.8	30.7	23.9	29.5	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.8	29.7	22.8	18.4	28.7	22.2	17.8	28.1	81.5	32.1	77.8	30.2	73.9	30.0	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.8	9.2	DB	-25.0	36.5	2.2	1.5	-26.6	37.6	-27.9	38.5	-29.2	39.4	-30.8	40.5	(4)
(5)			WB	-25.4	27.4	2.2	0.8	-27.0	27.9	-28.2	28.4	-29.5	28.8	-31.0	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-5.0	-2.9	3.6	9.8	15.9	21.6	24.3	22.5	18.6	10.9	3.2	-3.4	(6)	
	DBStd	11.60	6.78	6.84	6.63	5.48	4.98	3.89	3.21	3.17	4.85	5.34	5.80	6.12	(7)	
	HDD10.0	1805	465	362	217	69	8	0	0	2	52	213	417		(8)	
	HDD18.3	3584	723	595	458	261	108	14	2	4	54	236	453	675	(9)	
	CDD10.0	1803	0	1	17	62	191	348	444	388	261	81	10	0	(10)	
	CDD18.3	542	0	0	1	4	33	113	187	134	63	7	0	0	(11)	
	CDH23.3	5704	0	0	13	77	394	1186	1983	1256	696	96	3	0	(12)	
(13)	CDH26.7	2211	0	0	2	20	131	451	858	471	250	27	0	0	(13)	
(14)	Wind	WSAvg	4.1	4.2	4.4	4.8	5.2	4.5	3.9	3.3	3.1	3.6	4.2	4.1	(14)	
(15) Precipitation	PrecAvg	718	17	19	40	72	110	122	88	79	61	54	34	22	(15)	
	PrecMax	963	35	40	93	116	215	284	230	244	149	163	92	71	(16)	
	PrecMin	504	4	2	0	10	34	41	24	18	13	4	1	1	(17)	
	PrecStd	123	8	10	26	29	50	60	53	51	37	40	25	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	18.0	25.1	29.0	32.9	35.1	37.1	35.9	33.8	30.1	22.7	13.9	(19)	
		MCWB	7.0	9.7	13.5	17.0	20.1	23.2	24.6	23.8	21.9	19.2	13.4	8.6	(20)	
	2%	DB	10.0	12.9	21.3	25.9	30.0	32.8	34.1	32.9	32.1	26.1	18.8	11.0	(21)	
		MCWB	5.0	7.3	12.9	15.3	19.5	22.9	24.6	24.2	21.7	16.7	11.6	6.4	(22)	
	5%	DB	7.1	9.8	17.5	22.4	27.6	31.2	32.7	31.3	29.4	22.8	16.0	7.7	(23)	
		MCWB	3.2	5.5	11.0	13.9	18.3	22.2	24.3	23.7	20.9	14.8	9.7	3.9	(24)	
	10%	DB	3.9	7.0	14.0	19.1	25.2	29.0	31.3	29.0	27.5	20.6	12.7	5.1	(25)	
		MCWB	1.4	3.6	8.6	12.2	17.2	21.1	23.8	22.7	19.2	14.1	7.7	2.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	11.1	16.6	18.4	22.6	25.7	27.1	26.7	24.1	20.6	14.8	9.9	(27)	
		MCDB	14.1	16.3	22.5	26.1	29.5	32.0	33.3	32.2	30.8	27.5	19.9	13.4	(28)	
	2%	WB	5.3	7.9	14.0	16.4	20.8	24.2	26.0	25.3	22.9	18.4	12.5	6.2	(29)	
		MCDB	9.6	12.6	19.0	24.0	28.1	30.4	32.3	30.9	29.3	23.4	18.1	10.5	(30)	
	5%	WB	3.3	5.5	11.5	14.7	19.4	23.2	25.2	24.3	21.7	16.3	10.2	4.1	(31)	
		MCDB	6.5	9.6	17.6	21.3	25.7	29.3	31.4	29.7	27.7	21.2	15.3	7.5	(32)	
	10%	WB	1.6	3.5	9.0	12.9	17.9	22.0	24.2	23.3	20.5	14.4	7.9	2.3	(33)	
		MCDB	4.0	6.6	14.2	18.9	24.0	27.9	30.1	28.3	26.4	19.5	12.5	5.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.1	12.3	12.9	12.3	12.0	11.5	11.6	13.6	13.3	12.4	10.7	(35)	
		MCDBR	15.3	15.8	18.3	17.6	15.9	13.9	12.9	13.6	15.1	17.6	18.2	15.0	(36)	
	5% WB	MCWBR	11.2	11.0	11.3	9.9	7.9	6.8	6.5	7.0	7.3	9.8	11.6	11.0	(37)	
		MCWBR	14.5	15.1	16.6	15.4	13.8	12.0	11.4	11.8	13.0	13.9	16.9	14.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.279	0.292	0.320	0.373	0.393	0.406	0.413	0.411	0.373	0.327	0.288	0.278	(40)	
	taud		2.460	2.457	2.415	2.281	2.302	2.323	2.309	2.309	2.370	2.486	2.510	2.470	(41)	
	Ebn at Noon		884	925	931	896	883	869	859	850	861	871	867	854	(42)	
	Edh at Noon		75	88	104	128	129	126	127	123	108	85	71	69	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.10	3.00	3.93	5.03	5.86	6.65	6.61	5.68	4.62	3.28	2.32	1.73	(44)	
	RadStd		0.18	0.19	0.32	0.30	0.39	0.45	0.36	0.39	0.29	0.35	0.16	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page